

## Education

- 2024–2027 **UPV Universitat Politècnica de València**, *Department of Telecommunication Engineering, Valencia, Spain*, Ph.D. in Electrical Engineering and Communication Networks  
**Research topic:** Integration of Terrestrial and Non-terrestrial Networks for Aerial and Ground Wireless Connectivity (Using Deep Reinforcement Learning).  
**Advisor:** Dr. David López-Pérez  
**GPA:** 9/10  
**Ph.D. Courses:**  
  - Next-Generation Wireless Networks — Grade: 9/10
  - Next-Generation Wireless Systems Applications — Grade: 9/10
- 2011–2014 **University of Tehran**, *Department of Electrical and Computer Engineering, Tehran, Iran*, M.Sc. in Electrical Engineering (Communication Systems)  
**Thesis:** Parallel Cooperative Spectrum Sensing in Cognitive Radio Networks.  
**Advisor:** Dr. Amir Masoud Rabie  
**GPA:** 14.56/20 (2.912/4)
- 2006–2011 **University of Tabriz**, *Department of Electrical and Computer Engineering, Tabriz, Iran*, B.Sc. in Electrical Engineering (Electronics)  
**Thesis:** Energy Optimization in Triac and Diac Based Light Dimmers.  
**Advisor:** Dr. Tohid Alizadeh  
**GPA:** 15.02/20 (3.008/4)

## Research Interests

Artificial Intelligence & Machine Learning (CNNs, Federated Learning, Deep Reinforcement Learning (Proximal Policy Optimization, Twin Delayed Deep Deterministic Policy Gradient)), ChatGPT and Large Language Models, Optimization, Wireless Communications and MIMO Systems.

## Publications

### Journal Articles

1. **Akhtarshenas, A.**, Vahedifar, M. A., Ayoobi, N., Maham, B., Alizadeh, T., et al. (2024). “*Federated Learning: A Cutting-Edge Survey of the Latest Advancements and Applications*.” *Computer Communications*, 228, 107964.
2. **Akhtarshenas, A.**, Toosi, R. (2022). “*An Open-Set Framework for Underwater Image Classification Using Autoencoders*.” *SN Applied Sciences*, 4(8), 229.

### Conference Papers

1. **Akhtarshenas, A.**, Ayoobi, N., López-Pérez, D., Toosi, R., Amoozadeh, M. (2024). “*CNN Autoencoder Resizer: A Power-Efficient LoS/NLoS Detector in MIMO-Enabled UAV Networks*.” In *2024 IEEE 35th International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*.
2. Alizadeh, T., **Akhtarshenas, A.**, Karimi, N., Soleimani, H., Alizadeh, A. (2024). “*PLC Programming Instruction: A Review of Popular Software Platforms for Enhanced Learner Outcomes*.” In *2024 9th International Conference on Mechatronics Engineering (ICOM)*, pp. 76–82.
3. **Akhtarshenas, A.**, Toosi, R., López-Pérez, D., Alizadeh, T., Hosseini, A. (2025). “*Efficient Malicious UAV Detection Using Autoencoder-TSMamba Integration*.” In *Iberian Conference on Pattern Recognition and Image Analysis*, pp. 309–320.

4. Vahedifar, M. A., **Akhtarshenas, A.**, RafatPanah, M. M., Sabbaghian, M. (2025). "Shapley-Based Data Valuation with Mutual Information: A Key to Modified K-Nearest Neighbors." In *2025 IEEE 35th International Workshop on Machine Learning for Signal Processing*.
5. Ibáñez, M. R., **Akhtarshenas, A.**, López-Pérez, D., Geraci, G. (2025). "Optimizing UAV Aerial Base Station Flights Using DRL-Based Proximal Policy Optimization." In *2025 IEEE International Conference on Communications Workshops (ICC Workshops)*.

#### Submitted / Under Review

1. Svistunov, G., **Akhtarshenas, A.**, López-Pérez, D., Giordani, M., Geraci, G., Yanikomeroğlu, H. "Bridging Earth and Space: A Survey on HAPS for Non-Terrestrial Networks." Submitted to *IEEE Communications Surveys & Tutorials*, arXiv:2510.19731, 2025.
2. **Akhtarshenas, A.**, Ibáñez, M. R., Bernabé, M., López-Pérez, D., Debbah, M. "Centralized PPO-Based DRL for Multi-UAV-BS Positioning and Trajectory Optimization in Disaster Response Networks." Submitted to *IEEE Transactions on Machine Learning in Communications and Networking*, 2026.
3. **Akhtarshenas, A.**, Mostafavi Tehrani, S. H., Toosi, R., Alizadeh, T., López-Pérez, D. "A Comprehensive Survey of Reinforcement Learning Algorithms: Taxonomy, Mathematical Foundations, and Practical Insights." Submitted to *CAAI Transactions on Intelligence Technology*, 2026.
4. **Akhtarshenas, A.**, Svistunov, G., Bernabé, M., Zheng, K., López-Pérez, D. "PPO-Based Dynamic Positioning of HAPS-BS in Wind-Disturbed Stratospheric Maritime Networks." Submitted to *IEEE GLOBECOM 2026*, arXiv:2605.05240, 2026.
5. **Akhtarshenas, A.**, Svistunov, G., Zheng, K., López-Pérez, D. "Wind-Resilient Trajectory Optimization for UAV-BS Networks: TD3 for Continuous Service Availability." Submitted to *IEEE GLOBECOM 2026*, arXiv:2606.18556, 2026.
6. Spampinato, L., Togola, M., Bernabé, M., **Akhtarshenas, A.**, Amorosa, L. M., López-Pérez, D. "Indirect Estimation of SINR via SSB and CSI-RS RSRP in 5G NR." Submitted, 2026.

#### Preprints / arXiv

1. **Akhtarshenas, A.**, Dini, A., Ayoobi, N. (2025). "ChatGPT or a Silent Everywhere Helper: A Survey of Large Language Models." arXiv:2503.17403.

### Profiles

LinkedIn [linkedin.com/in/azim-akhtarshenas](https://www.linkedin.com/in/azim-akhtarshenas)  
 Google Scholar [scholar.google.com/citations?user=5eCLyxwAAAAJ](https://scholar.google.com/citations?user=5eCLyxwAAAAJ)  
 GitHub [github.com/aakhtarshenas](https://github.com/aakhtarshenas)

### Technical Skills

Programming Python, MATLAB, C, Verilog  
 Machine Learning & Deep Learning PyTorch, TensorFlow, TorchRL, NumPy, Pandas, Scikit-learn  
 Experiment Tracking & MLOps MLflow, Weights & Biases (W&B)  
 Simulation & Engineering Tools Simulink, LabVIEW, ModelSim, Proteus, PSpice, HSpice, CodeVision

Wireless & Hardware Platforms USRP boards, AVR microcontrollers, TI DSPs

Productivity Tools L<sup>A</sup>T<sub>E</sub>X, MS Office Suite (Word, Excel, PowerPoint), Visio

### Personal Skills

Excellent communication and interpersonal skills; leadership, management, and presentation skills; strong decision-making ability; effective planning and prioritization.

## PhD Summer Schools

- 2024 **Summer School on Advanced Topics in 5G/6G Communications**, Instructor: Dr. David López-Pérez, Universidad Internacional Menéndez Pelayo, Cuenca, Spain, June 1–4, 2024
- 2025 **Summer School on Advanced Topics in 5G/6G Communications**, Instructor: Dr. David López-Pérez, Universidad Internacional Menéndez Pelayo, Cuenca, Spain, June 1–4, 2025
- 2025 **IEEE/DEI Summer Ph.D. School of Information Engineering**, Instructor: Dr. Marco Giordani, Università degli Studi di Padova, Brixen, Italy, July 6–10, 2025

## Experiences

### Teaching Experience

- Winter 2009 **Linear Control Systems**, Instructor: Dr. Tohid Alizadeh
- Winter 2015 **Communication (2)**, Instructor: Dr. Amir Masoud Rabie
- Spring 2016 **Engineering Probability and Statistics**, Instructor: Dr. Amir Masoud Rabie
- Fall 2016 **Advanced Theory of Communication**, Instructor: Dr. Amir Masoud Rabie
- Spring 2017 **Signals and Systems**, Instructor: Dr. Amir Masoud Rabie
- Winter 2017 **Data Network**, Instructor: Dr. Vahid Shah-Mansouri
- Spring 2018 **Advanced Theory of Communication**, Instructor: Dr. Amir Masoud Rabie
- Fall 2018 **Engineering Probability and Statistics**, Instructor: Dr. Amir Masoud Rabie
- Spring 2020 **Signals and Systems**, Instructor: Dr. Amir Masoud Rabie
- Winter 2020 **Stochastic Processing & Random Variables**, Instructor: Dr. Amir Masoud Rabie
- Fall 2021 **Signals and Systems**, Instructor: Dr. Amir Masoud Rabie
- Winter 2021 **Stochastic Processing & Random Variables**, Instructor: Dr. Amir Masoud Rabie

### Work Experience

- 2019 Implementation of the Internet network at Eslamshahr Azad University
- 2016–2017 Military service as technical staff at the communication room of Kooy Dormitory, University of Tehran
- 2021 Implementation of a wireless communication network under Voice over Internet Protocol (VoIP) at the College of Mechanical Engineering, University of Tehran

## Awards & Honors

- 2009 Ranked 719th (regional rank) and 7200th (national rank) in the National Universities Entrance Examination among more than 308,000 contestants.
- 2013 Ranked 125th in Communications Engineering in the Graduate Entrance Examination among more than 1,500 contestants.
- 2016–2017 Ranked 1st as the communication systems operator in the dormitories of the University of Tehran as part of mandatory military service among more than 100 applicants.

## Course Projects

- Fall 2010 **Energy optimization in infrared control**, *Microprocessors*
- Investigated the effects of obstacles in infrared control and methods for efficient operation.
  - Implemented energy-saving hardware solutions.
- Fall 2013 **Combined Spatial Multiplexing/STBC Systems**, *Coding Theory*
- Analyzed the paper “Improved Linear Group Detection for Combined Spatial Multiplexing/STBC Systems.”
  - Studied group-based and direct detection techniques for STBC streams.
  - Compared complexity and performance of both approaches.
- Fall 2013 **Sensor selection via convex optimization**, *Convex Optimization*
- Studied capacity maximization under different power constraints in dual-hop cooperative communication networks.
  - Analyzed “Antenna Placement Optimization for Distributed Antenna Systems.”
- Fall 2013 **Automated Bayesian system identification**, *System Identification*
- Analyzed “Automated Bayesian system identification in NARX Model.”
  - Designed and evaluated Bayes, linear, Parzen window, and KNN classifiers.
- Fall 2012 **Dynamic Decode and Forward (DDF)**, *Information Theory*
- Analyzed “Network-level Cooperation for a Multiple Channel via Dynamic Decode and Forward.”
  - Studied the effect of DDF and superposition coding on throughput and queue stability.

## Skills

### Language Skills

|         |                                                                                                |
|---------|------------------------------------------------------------------------------------------------|
| Turkish | Native                                                                                         |
| Persian | Native                                                                                         |
| English | Fluent; TOEFL: R 22, L 21, S 24, W 22. GRE: Verbal 144/170, Quantitative 165/170, Writing 3/6. |
| Spanish | Fluent                                                                                         |
| German  | Familiar                                                                                       |
| Arabic  | Familiar                                                                                       |

## Graduated & Relevant Courses

### Online

2018–2019 Deep Learning (Coursera), Reinforcement Learning (University College London)

### Audited

2020 Advanced Algebra (University of Tehran)

### Classroom

2011–2014 Convex Optimization, Advanced Digital Communication, Stochastic Processes and Random Variables, Coding Theory, Information Theory, Data Networks, System Identification

## Paper Editor

- 2020 **Pervasive and Mobile Computing Journal**, “*The Game Selection Model for the Payoff Strategy Optimization of Mobile CrowdSensing Task*”
- 2020 **Pervasive and Mobile Computing Journal**, “*TOTP Moving Target Defense for sensitive network services*”

## References

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